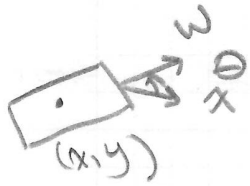
Basic Motion Models

I Unicycle Model

$$m = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix}$$



$$W = \begin{bmatrix} \cos\theta \\ \sin\theta \\ 0 \end{bmatrix}$$

$$\Rightarrow W \times m = \begin{bmatrix} 0 \\ 0 \\ c\theta\dot{y} - s\theta\dot{x} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \cos\theta\,\dot{y} - \sin\theta\,\dot{x} = 0$$

$$\Rightarrow \dot{y} = v\sin\theta \qquad \dot{x} = v\cos\theta \qquad \dot{\theta} = \omega$$

2 DOF because v & ω are chosen

$$\Rightarrow \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} v\cos\theta \\ v\sin\theta \\ \omega \end{bmatrix} \quad \text{where } v = \omega \cdot r$$

II Differential Drive

$V_l = \mu_l r \qquad V_r = \mu_r r$

Assume (x,y) is in the middle of robot

θ is defined from x-axis

- Be careful w/ signs μ_l and μ_r. One will be positive and the other negative because we define z as coming out of the robot.

$\dot{x} = v\cos\theta \qquad \dot{y} = v\sin\theta \qquad v \in \mathbb{R}$

$\Rightarrow v = \frac{v_r}{2} + \frac{v_l}{2}$

$\dot{\theta} = u_r + u_l = \frac{v_r}{L} - \frac{v_l}{L}$

$$\Rightarrow \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} \frac{v_r - v_l}{2} \cos\theta \\ \frac{v_r - v_l}{2} \sin\theta \\ \frac{v_r - v_l}{L} \end{bmatrix}$$

where

$v_r = u_r r$

$v_l = u_l r$

III Bicycle Model (car model)

ϕ : steering angle

$\dot{x} = v\cos\theta$

$\dot{y} = v\sin\theta$

$\dot{\theta} = \frac{v}{L}\tan\phi$